



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 5: Logistics Systems

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- the basics of a production system from an applicative point of view; its significance in the design and development process of systems; Pull vs. Push production systems; Mass, Batch, Continuous and intermittent production systems; Some vital processes in a production system and finally, a production system contextualised in terms of the semester project with rudimentary tips on how to address selected sections of the project rubric.

Expected Learning Outcomes for the Current Lecture

At the end of this lecture you should:

- Understand the basic concepts about logistic systems including the broad definition
- Understand the basic principles of logistics management
- Understand and be able to apply, analyse, evaluate and create customised attributes of a logistic subsystem within a project or business system.
- Understand and be able to illustrate the operations of a logistic system from a point of view of the semester project

Expected Learning Outcomes for the Current Lecture.....Contd.

- Be able to explain the interfacing behaviour of a logistic system with other systems and sub-systems up until the end of their life cycle.

BLOOM'S TAXONOMY & EXPECTED LEVELS of COGNITION

BLOOM'S TAXONOMY

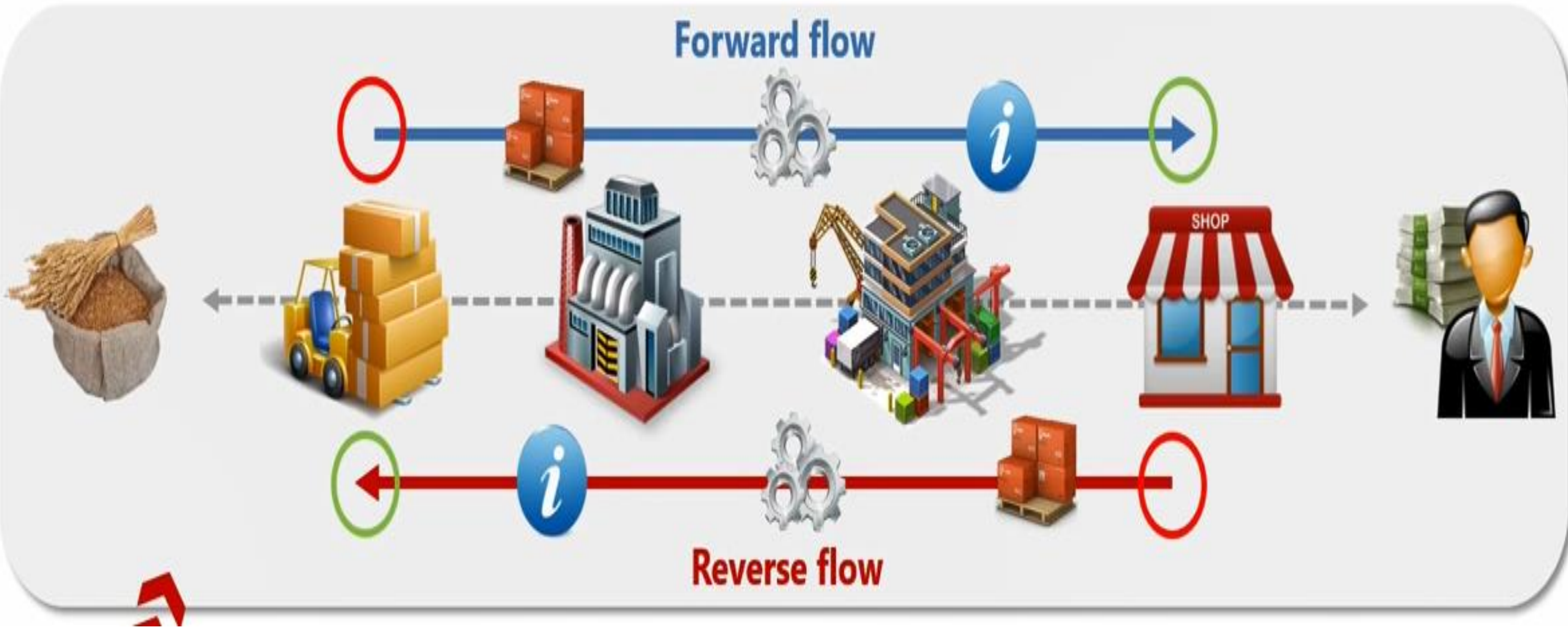


Presentation Outline

- Definition of Logistics Systems
- Principles of Logistics Management
- Some Characteristics of Logistics Systems
- Operations of Logistics Systems
- Logistics Systems Interfaced with other Systems/Sub-systems

Definition of Logistics

Logistics is often defined as that part of a system's operations that focuses on planning, implementation and controls for efficient, effective forward and reverse flow and storage of goods, services, and related information between the point of origin and the point of consumption in order to meet customers' requests.



Forward and Reverse Logistics

Definition of Logistics Systems...Contd

Logistics is also defined as the art and science of obtaining and distributing materials and products at the right location, quantity and time.

Logistics is also referred to as the movement of materials along the phases of a supply chain

Internal & External Logistics

Inbound logistics is concerned with purchases and inflow movement of materials, parts, and/or finished inventory from suppliers to manufacturing or assembly plants and warehouses.

Outbound logistics is concerned with the storage and movement of final products and/or information flow from the end of the production line to the end user.

Principles of Logistics Management

The seven R's of Logistics Systems include:

- Right Product
- Right Quantity
- Right Condition
- Right Place
- Right Time
- Right Customer
- Right Price

Principles of Logistics Management



Right Product



Right Quantity



Right Condition



Right Place



Right Time



Right Customer



Right Price

Some Characteristics of Logistics Systems

Respond Rapidly

Control Inventory Variance

Monitor Inventory Expenses

Consolidate Shipments

Sustain Products Quality

Supports Products Life Cycle

Conduct Reverse Logistics

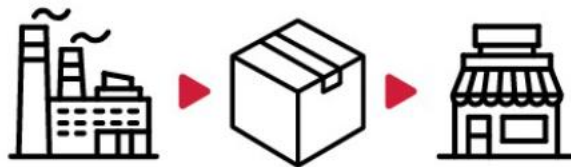
Operations of Logistics Systems



1st Party Logistics

The 1st party logistics is often associated with a firm that has its own cargo, freight and can transport goods and merchandise from one point to another.

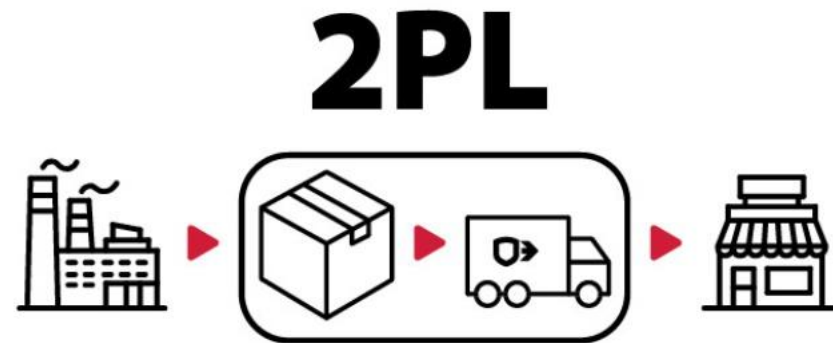
1PL



2nd Party Logistics

A 2nd party logistics involves transportation of goods from one end of a supply chain to a destination of need using rails, road, sea and/or air. They include the asset-based carriers using ships of own lease and airlines that they are contracted with. They are mainly used for international transportation of heavy and wholesale goods and for trading purpose as well.

2nd Party Logistics



3rd Party Logistics

The 3rd party logistics is primarily concerned with the transportation and delivery of different products but also includes various types of additional services as well. The functions of the 3PL include warehousing, terminal operations, customs brokerage, supply chain management, logistics IT software products and analysis services, for tracking and tracing the delivery status of different products amongst others. They specialize in domestic and offshore warehousing.

3rd Party Logistics



1st, 2nd & 3rd Party Logistics Facilitation



4th Party Logistics

The 4th party logistics is a relatively new concept in business systems logistics. Usually, it involves employing an overseer referred to as a Logistic Service Provider to aid with the management of an entire supply chain of a company.

4th Party Logistics

4PL



5th Party Logistics

The 5th party logistics providers are known as logistics aggregators. They aggregate the demands of the 3PL and others into bulk volume for getting better rates with different types of airlines and shipping companies. This type of logistics is not asset based. It usually works seamlessly across all disciplines.

5th Party Logistics

5PL



6th Party Logistics

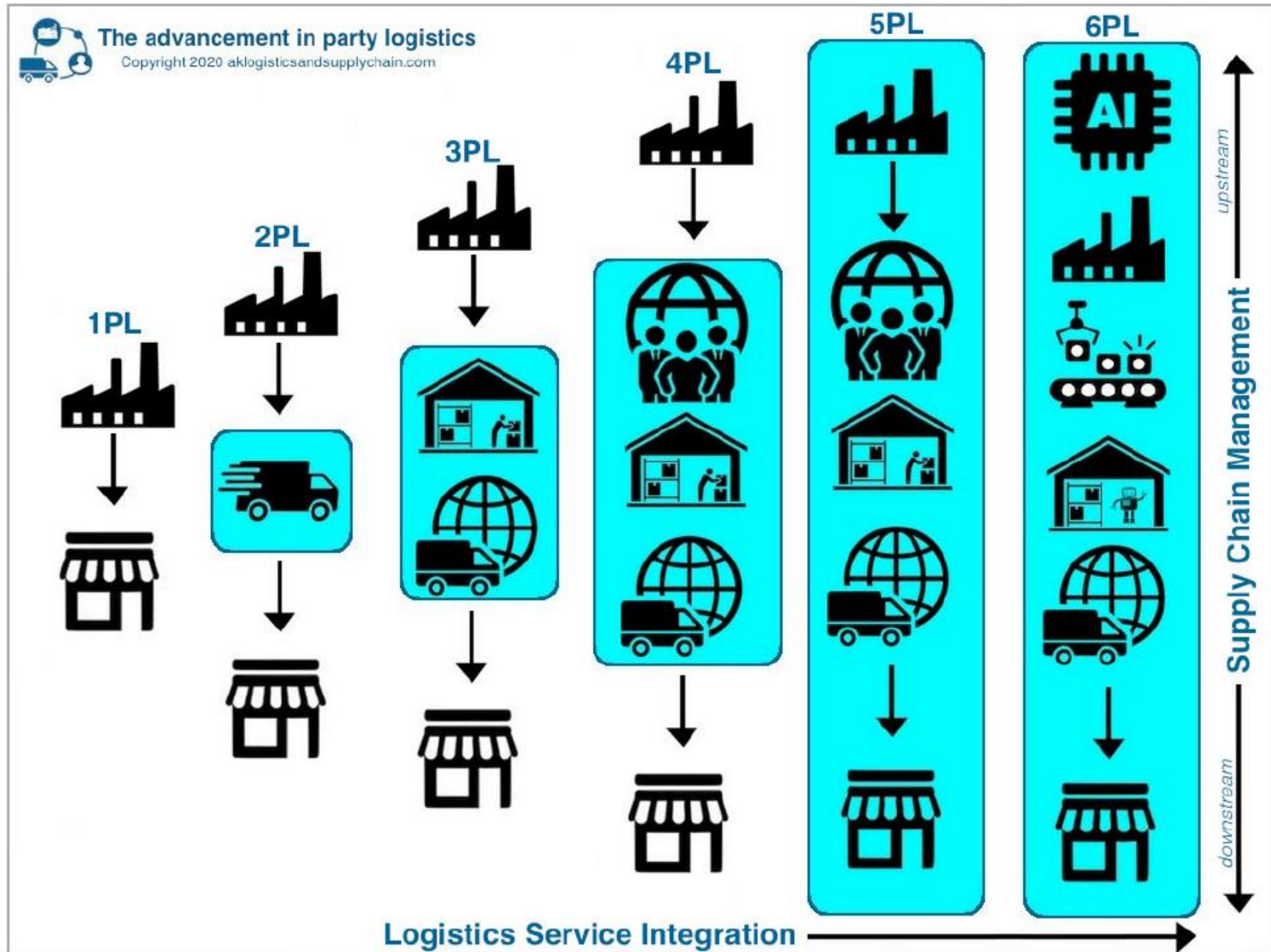
6PL





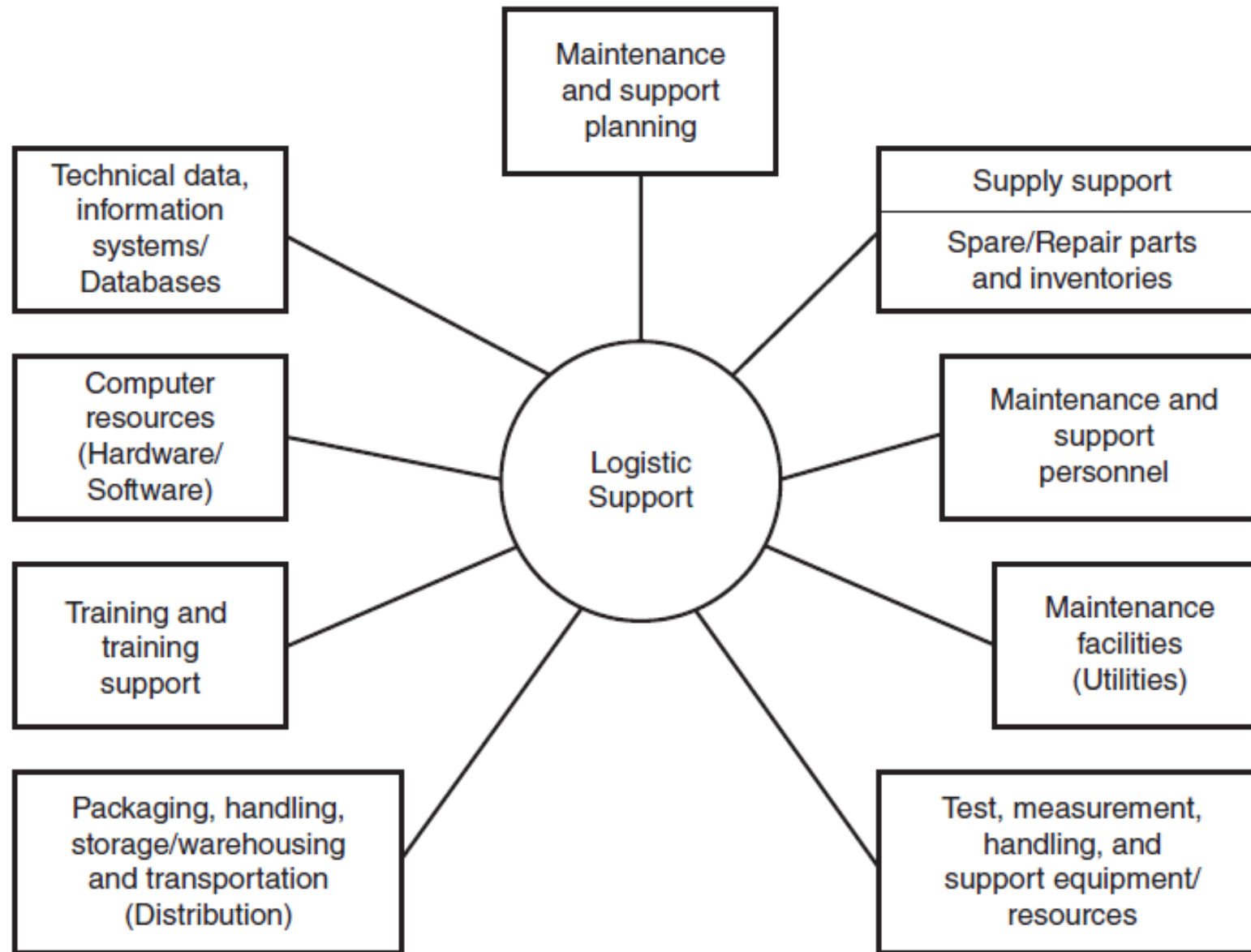
The advancement in party logistics

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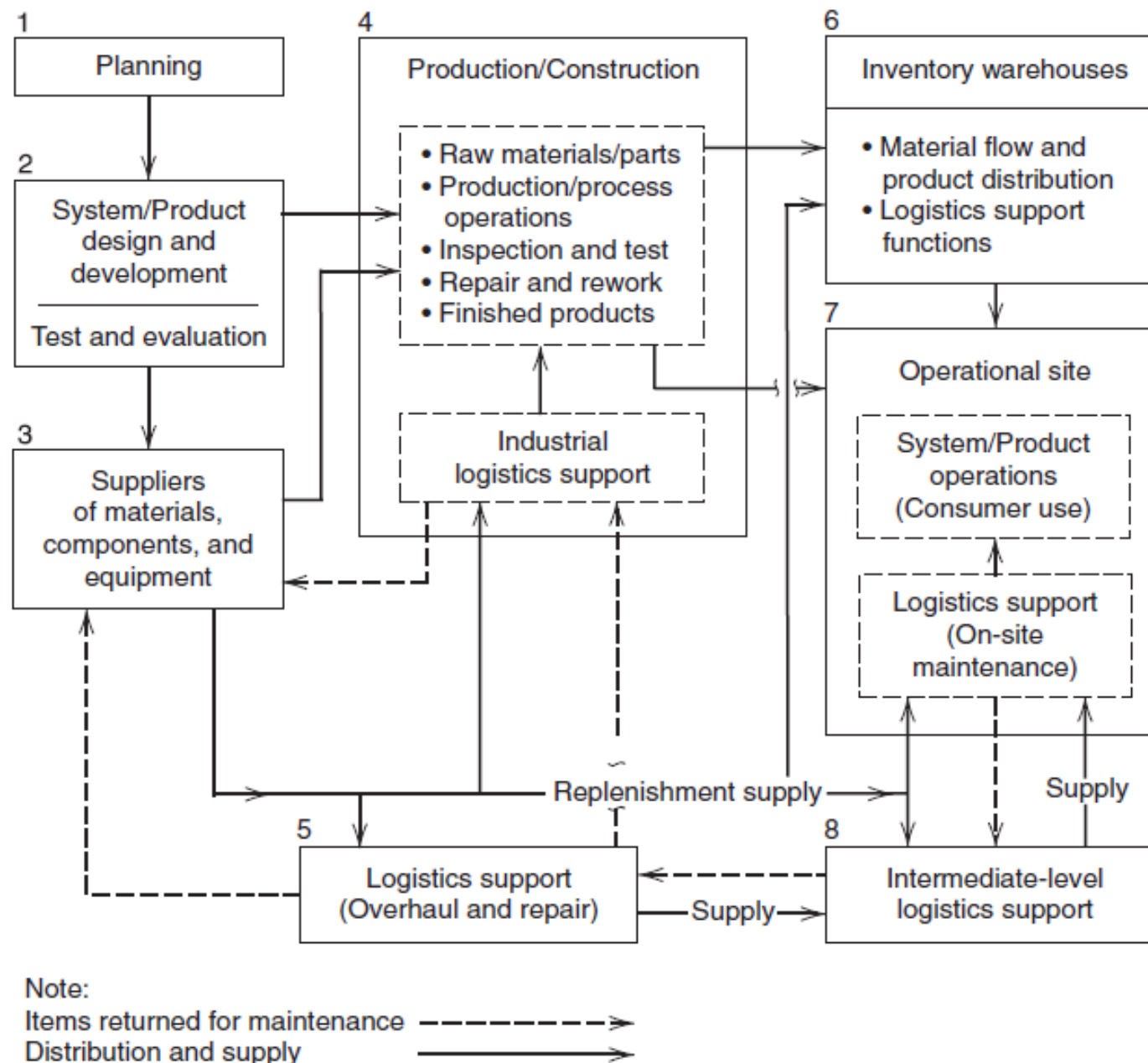


Summary Sheet 1st to 6th Party Logistic System

Logistics Systems Interfaced with a System, Production and Marketing Sub-Systems



Logistic System Interfaced with the entire System

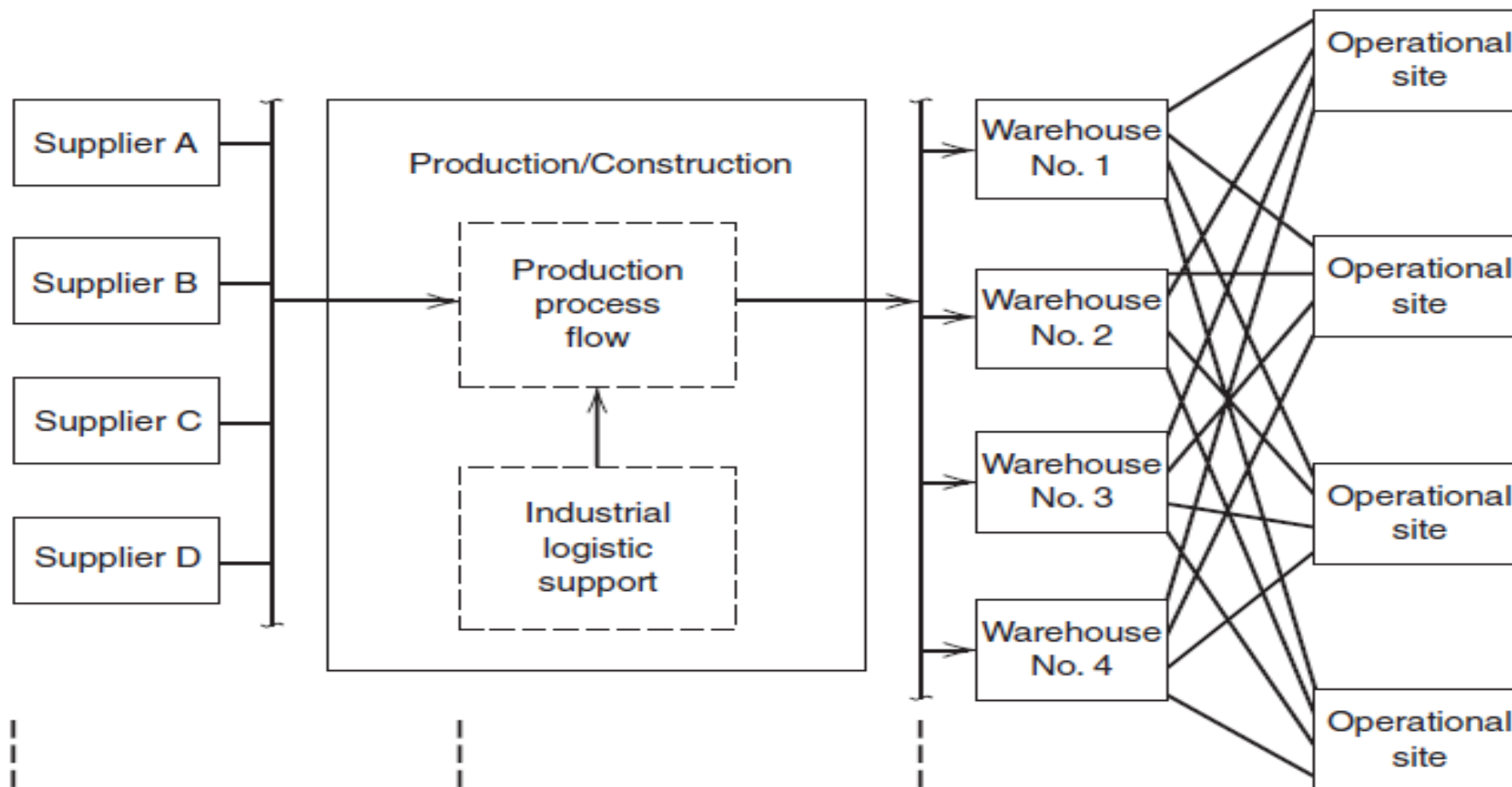


Logistics and Production System Interfaced

Logistics-Production Interface

How does each of these scenarios affect the logistics decision factors?

- Longer production runs (e.g. 1 000 units at a time instead of 20) to reduce setup costs. A possible make to stock policy (a Push systems).
- Just-In-Time production systems (Pull systems)
- Choosing to use licensed instead of open source software for service development
- Location of production processes



PHYSICAL SUPPLY

- Demand forecasting
- Order processing
- Procurement
- Inventory management
- Transportation/flow
- Information flow

MANUFACTURING

- Production planning
- Purchasing
- Materials handling
- Inventory management
- Packaging/shipping
- Information flow

PHYSICAL DISTRIBUTION

- Demand forecasting
- Order processing
- Inventory management
- Transportation/traffic
- Customer service
- Information flow

Logistics-Marketing Interface

How does each of these scenarios affect the logistics decision factors?

- A half-price sale promotion for retail goods
- A bulk price discount offered to customers
- Free overnight door-to-door delivery for online purchases
- Returns policies and warranty periods
- On-time delivery promises

Learning Outcomes Evaluated

- Do you now have some level of confidence when it comes to expressing yourself in the dynamic world of logistics?
- Can you now define using your own words, and explain the basic concepts about logistic systems?
- What about the basic principles of logistics management with the seven R's can you discuss this and apply to your business?
- Can you at a basic level, discuss the operations of a logistics system? Including being able to analyse, evaluate and create customised attributes of a logistic subsystem within a project or business system.

Learning Outcomes Evaluated

- What about being able to explain the interfacing behaviour of a logistic system with other systems and sub-systems up until the end of their life cycle? Can you tell a story around this?
- If your response to the above tracking queries is “yes”, count yourself sorted with related assessments in this module. Also, you can comfortably start your own logistics business or apply for a logistics centric job.

For further reading: Sections covered in the e-book include:

- **System Engineering Management, Fifth Edition, Benjamin S. Blanchard & John E. Blyler,**
 - Chapter 1 sections 1.4, 1.4.3 and 1.4.4
 - Chapter 3 section 3.4.8

Additional resources were randomly sourced from:

- Systems Engineering Principles and Practice, Second Edition; Author: Alexander Kossiakoff et al. Chapter 15 section 15.3
- <https://www.shopify.co.za/encyclopedia/logistics>
- <https://ithinklogistics.com/blog/types-of-logistics-its-difference/>
- <https://www.linkedin.com/pulse/decoding-logistics-providers-unveiling-differences/>

The End

Next lecture

- Unit 6: Marketing and customer service



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